

Today's Agenda:-

Starting 7:05

1. Special Index
2. Number of even no's in given range.
3. Leaders in an Array.

Quiz 1:-

0 1 2 3 4
2, 4, 3, 1, 5
pFo = {0, 4, 4, 5, 5}

$$pFo[i] = pFo[i-1] + \begin{cases} 0 & i \text{ is even} \\ arr[i] & i \text{ is odd} \end{cases}$$

Q: 1. Given an array of size N.
Count the no. of special indices in the array.

Special indices are those after removing which, sum of all even indexed elements

=

odd indexed elements

ex:-

{⁰4, ¹3, ²2, ³7, ⁴6, ⁵-2}

i	A[i]					Se	So
	0	1	2	3	4		
0	3	2	7	6	-2	8	8
1	4	2	7	6	-2	9	8
2	4	3	7	6	-2	9	9
3	4	3	2	6	-2	4	9
4	4	3	2	7	-2	4	10
5	4	3	2	7	6	12	10

ans = 2.

Ans: 2

0 1 2 3 4
2 4, 1, 3, 7, 10
4 1 7 10
3

11.

i	A[i]				Se	So
	0	1	2	3		
0	1	3	7	10	8	13
1	4	3	7	10	11	13
2	4	1	7	10	11	11
3	4	1	3	10	7	11
4	4	1	3	7	7	8

ans = 1.

B.F. Idea:-

Try out all possible indices by storing the remaining elements in an array of size $N-1$.

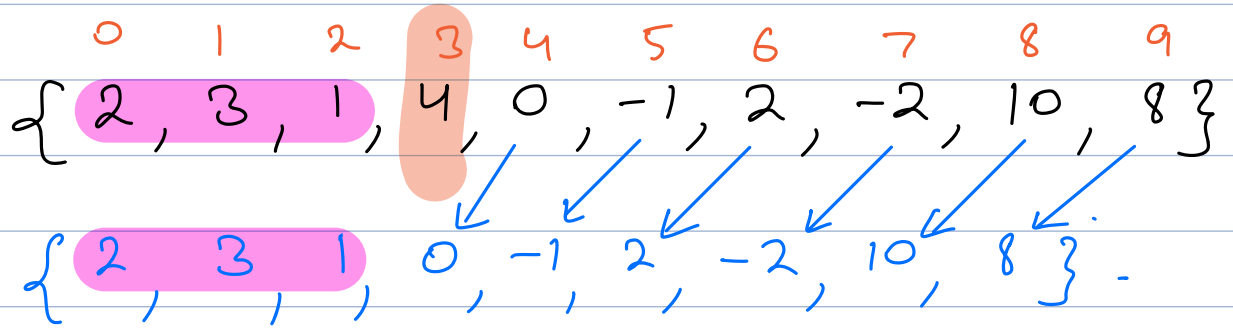
Iterate on that array to calculate Se & So.

If equal, increment the count.

T.C $\rightarrow O(N^2)$.

Idea 2:-

Q:- 3



ans = 15

Sum of all odd indices after removing
 j^{th} index

= Sum of all odd indices
[0, i-1]

+

Sum of all even indices
[i+1, N-1];

$$\text{Sum}_{\text{ODD}} = \text{Sum}_{\text{ODD}} [0, i-1] + \text{Sum}_{\text{EVEN}} [i+1, N-1]$$

$$= \text{pSumO}[i-1] + \text{pSumE}[N-1] - \text{pSumE}[i+1-1];$$

Q:-4.

0 1 2 3 4 5 6 7 8 9
{ 2, 3, 1, 4, 0, -1, 2, -2, 10, 8 }

∴ { 2, 3, 1, 0, -1, 2, -2, 10, 8 }

10.

Sum of all ^{even} indices after removing
_{ith} index

= Sum of all ^{even} indices
[0, i-1]

+

Sum of all ^{odd} indices
[i+1, N-1];

Sum_{EVEN} = Sum_{EVEN} [0, i-1] + Sum_{ODD} [i+1, N-1]

= pSumE [i-1] + pSumO [N-1]
- pSumO [i+1-1];

Approach:-

1. Precalculate $psum_0$ & $psum_E$ arrays.
2. Iterate on every index from 0 to $N-1$ to find if it is a special index using above formula.
If yes, increment the count.

Code

```
int SpecialIndices (int[] arr) {
```

```
    int pSumE[] = pSumE(arr); N
```

```
    int pSumO[] = pSumO(arr); N
```

```
    int count = 0;
```

```
    int SE, SO;
```

```
    for (i = 0; i < N; i++) {
```

```
        if (i == 0) {
```

```
            SE = pSumO[N-1] - pSumO[0]; 0
```

```
            SO = pSumE[N-1] - pSumE[0];
```

```
        } else {
```

```
            SE = pSumO[N-1] - pSumO[i] + pSumE[i-1];
```

```
            SO = pSumE[N-1] - pSumE[i] + pSumO[i-1];
```

```
        if (SE == SO) {
```

```
            count++;
```

```
    }
```

```
    return count;
```

```
}
```

T.C $\rightarrow O(N)$

S.C $\rightarrow O(N)$

$\uparrow$

Because of
pSumE & pSumO
arrays.

8:45

Q:- 2.

Given an array of size N & Q queries. For every query, return the count of all even elements from s to e .

$arr[] \rightarrow \{-5, 10, 20, 40, 35, -2, -10\}$
 $pFEE \rightarrow \{0, 1, 2, 3, 3, 4, 5\}$

Queries

<u>s</u>	<u>e</u>	
0	4	3
1	2	2
2	5	3
3	3	1

$$pFEE[0] = 1 \text{ if } arr[0] \% 2 == 0 \\ \text{else } 0.$$

$$pFEE[i] = pFEE[i-1] + \begin{cases} 1 & arr[i] \% 2 == 0 \\ 0 & \text{!}(arr[i] \% 2 == 0) \end{cases}$$

$$[s, e] = pFEE[e] - pFEE[s-1] \\ \text{if } (s \neq 0) \\ \text{else} \\ pFEE[e];$$

Code to create pSumE array.

```
int[] pSumE (int[] arr) {
```

```
    int[] pSumE = new int [arr.size];
```

```
    if ( arr[0] % 2 == 0 ) {
```

```
        pSumE[0] = 1;
```

```
    } else {
```

```
        pSumE[0] = 0;
```

```
    for (i = 1 to N-1) {
```

```
        pSumE[i] = pSumE[i-1]
```

```
        + ( arr[i] % 2 == 0 ?
```

```
            1 : 0 );
```

```
    return pSumE;
```

```
}
```

void solve (int[] arr, int[][] queries) {

int pSum[] = pSumE(arr);

int s, e, sum;

for (q=0; q < queries.size(); q++) {

s = queries[q][0];

e = queries[q][1];

if (s == 0) {

sum = pSum[e];

} else {

sum = pSum[e] - pSum[s-1];

}

print (sum);

}

}

Q: 3. Given an integer array A containing N distinct integers. Find all the leaders in array A .
An element is a leader if it is strictly greater than all elements to its right side.

ex:- $\{16, 17, 4, 3, 5, 2\}$

ans = 3.

ex:- $\{5, 4\}$ ans = 2.

Q: 5 arr:- $\{10, 8, 6, 4, 2\}$

ans = 5.

Approach:-

For $A[i]$ to be a leader
 $\rightarrow A[i] > \text{All elements to right of it}$

$\downarrow$
 $A[i] > \text{max on Right}$
 $\downarrow$

$> \text{All on right:}$

Iterate from right to left & carry forward the max on right

i $\leftarrow$
0 1 2 3 4 5
 $\{16, 17, 4, 3, 5, 2\}$

Ans $\Rightarrow \{2, 5, 17\}$

maxOnRight = ~~-∞~~ ~~2~~ ~~5~~ 17

Q.13.

Code

ArrayList<> solve (int arr[]) {

int maxOnRight = INT_MIN;

ArrayList<int> ans = new ArrayList<>();

for (i = N-1; i >= 0; i--) {

if (arr[i] > maxOnRight) {

ans.add(arr[i]);

maxOnRight = arr[i];

}

return ans;

}

T.C $\rightarrow O(N)$.



